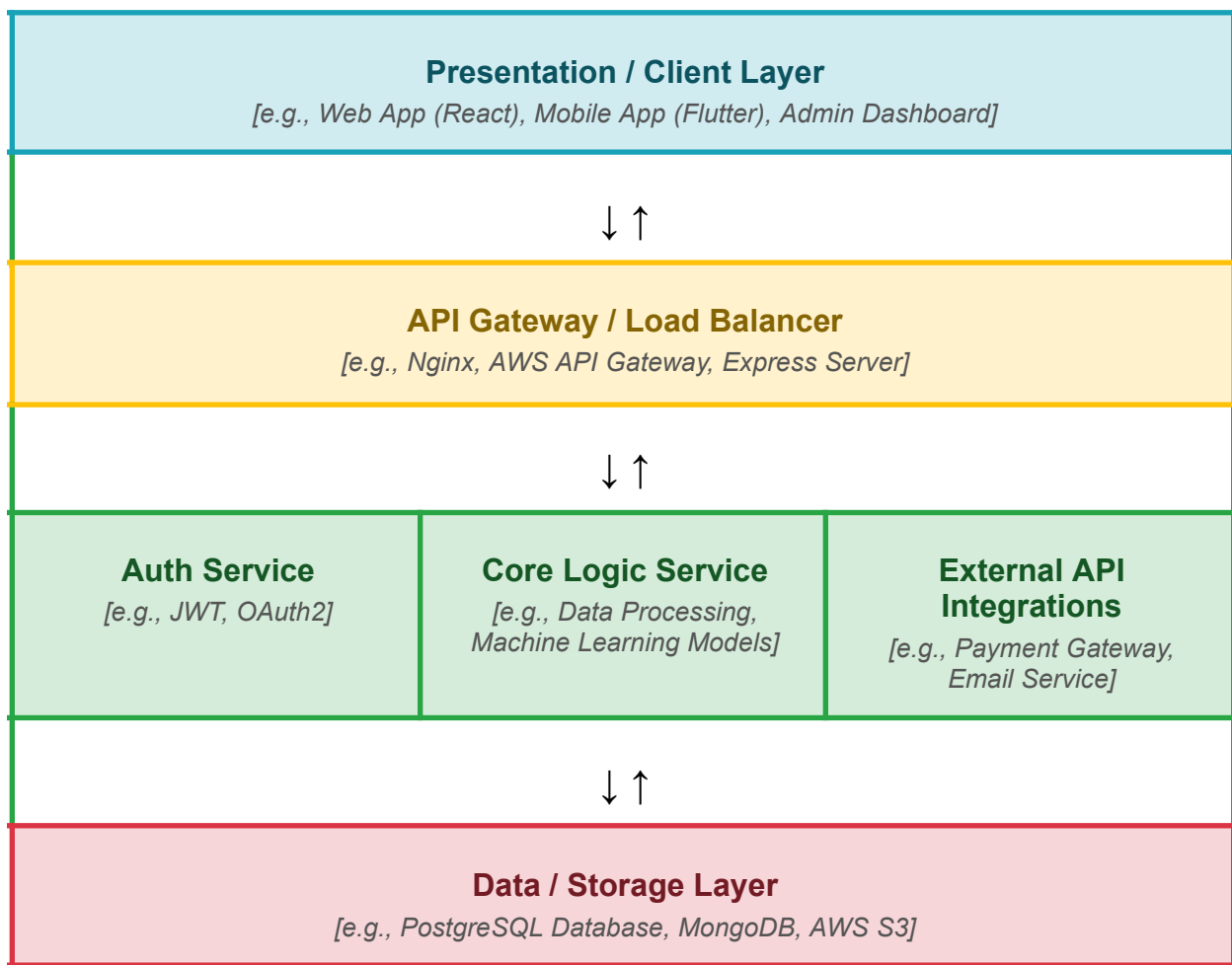


Solution Architecture

Date	28 th June 2026
Team ID	xxxxxxx
Project Name	Rising waters
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface where users enter environmental parameters and view flood prediction results.	HTML, CSS, JavaScript
API Gateway	Receives user requests from the frontend, forwards them to the prediction logic, and returns the prediction results.	Flask (Python)
Microservices	Handles data preprocessing, loads the trained ML model, performs flood prediction, and generates probability scores.	Python, Scikit-learn, XGBoost, Joblib

Database	Stores the historical flood dataset used for training and testing the machine learning model.	CSV Dataset (Flood_prediction.csv), Pandas
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